

MEHUL MITTAL

AI/ML Engineer | Deep Learning & GenAI Solutions

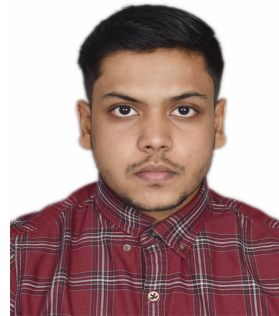
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Fürth, Bavaria (Open to Relocation)



Professional Summary

I'm an AI/ML Engineer who loves building production machine learning systems and working with deep learning and GenAI applications. I've worked on the full ML lifecycle, from initial research all the way to deployment, using tools like PyTorch, TensorFlow, and modern LLM frameworks. Over the years, I've built scalable neural networks, set up MLOps pipelines, and deployed AI solutions on Azure and AWS. My background covers computer vision, NLP, reinforcement learning, and transfer learning. I've published research and spent a lot of time on model optimization, hyperparameter tuning, and designing AI system architectures. What drives me is taking state-of-the-art ML research and turning it into real products that people can use.

Experience

ML Infrastructure Engineer (Working Student)

Robert Bosch | Mar 2025 – Sep 2025 | Abstatt, Germany

- Developed LLM solution using prompt engineering and fine-tuning to standardize technical diagrams, reducing manual review time by 60%.
- Built cloud ML infrastructure on AWS supporting distributed training and inference pipelines for computer vision and NLP models across global teams.
- Set up MLOps automation with GitHub Actions and Docker for continuous training, model versioning, and A/B testing.
- Optimized ML deployments on Lambda and EC2 through quantization and inference optimization, cutting costs by 25%.
- Worked with research teams on AutoML and neural architecture search experiments to establish hyperparameter tuning best practices.

AI Research Engineer (Working Student)

Machine Learning Lab, FAU | Mar 2024 – Apr 2025 | Erlangen, Germany

- Researched freeze of gait (FOG) detection in Parkinson's patients, developing deep learning models using CNNs, RNNs, and Transformers to identify FOG events from wearable sensor data in real-time.
- Applied signal processing techniques like wavelet transforms and FFT with feature engineering for real-time healthcare ML pipelines achieving sub-second latency.
- Developed transfer learning approaches combining self-supervised learning and domain adaptation for cross-subject model generalization, achieving state-of-the-art results.
- Co-authored IEEE-EMBS conference paper investigating subjective motor activity perception and gait patterns in Parkinson's patients using wearable sensors.

Lead AI/ML Engineer

Concentrix | Oct 2021 – Feb 2023 | Gurgaon, India

- Led development of production ML platform serving 200+ users with real-time predictions using ensemble methods, gradient boosting, and neural networks.
- Designed end-to-end ML pipelines with automated feature engineering, model training, hyperparameter tuning, and online learning capabilities.

- Built time-series forecasting and anomaly detection models delivering €2M ROI through optimization algorithms and reinforcement learning techniques.
- Set up model monitoring with drift detection, explainability tools like SHAP and LIME, and automated retraining pipelines to ensure production model performance.
- Mentored junior ML engineers on deep learning fundamentals, model architectures, and best practices for neural network training and optimization.

ML Engineering Intern

Mirrag AI | Sep 2021 – Nov 2021 | Mumbai, India

- Developed computer vision models for object detection and image classification using CNNs and transfer learning with pre-trained architectures (ResNet, EfficientNet, YOLO) achieving 85%+ accuracy on production datasets
- Implemented data augmentation strategies and preprocessing pipelines for image datasets improving model robustness by 20% and enabling generalization across diverse visual conditions
- Built MLOps foundation with experiment tracking (MLflow, Weights & Biases) enabling systematic evaluation of 50+ model configurations and reducing experimentation cycle time by 40%
- Established reproducible ML workflows with version control, data versioning (DVC), and containerization (Docker) accelerating team deployment process from weeks to days

Key AI/ML Projects

Neural Transfer Learning for Cross-Subject Generalization

Master's Thesis | May 2024 – Nov 2025

- Built end-to-end deep learning pipeline processing 1M+ samples with automated data augmentation, normalization, and quality validation using PyTorch Lightning.
- Implemented neural architectures like EEGNet, EEGConformer, ATCNet, and Vision Transformers with custom attention mechanisms and residual connections.
- Applied domain adaptation techniques including adversarial training, contrastive learning, and meta-learning for zero-shot generalization across subjects.
- Conducted ablation studies and hyperparameter optimization across model architectures to gain insights on transfer learning limitations in neural signal processing.
- Deployed MLOps infrastructure with MLflow for experiment tracking, model registry, and DVC for dataset versioning to enable reproducible research.

Healthcare Analytics with Large Language Models

Academic Capstone | Oct 2024 – Apr 2025

- Built GenAI pipeline integrating LLMs like GPT-4 and Claude with custom fine-tuning for medical domain using prompt engineering and few-shot learning.
- Developed RAG architecture with semantic search using vector embeddings and vector databases like Pinecone and ChromaDB for medical knowledge retrieval.
- Implemented automated monitoring for LLM drift detection, hallucination metrics, and response quality evaluation to ensure reliable AI outputs in regulated environments.
- Set up MLOps governance with MLflow 3.0 for experiment management, model versioning, and comprehensive audit trails meeting healthcare compliance standards.

Education

M.Sc. in Artificial Intelligence

Friedrich Alexander University Erlangen-Nuremberg | Apr 2023 – Jan 2026 (Completed)

Specialization: Deep Learning, Neural Networks, Computer Vision, NLP, Reinforcement Learning, MLOps, Probabilistic ML

Awarded FAU Graduation Scholarship (December 2025) for academic excellence in AI research

B.Tech in Information Technology

Guru Gobind Singh Indraprastha University | Aug 2018 – Jun 2022

Capstone: Team Lead for Distributed ML System Implementation

Technical Skills

Deep Learning & Neural Networks: PyTorch, TensorFlow, Keras, PyTorch Lightning, CNNs, RNNs, LSTMs, Transformers, Vision Transformers (ViT), Attention Mechanisms, ResNets, EfficientNet, BERT, GPT architectures

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Random Forests, SVM, Ensemble Methods, Time-Series Forecasting, Anomaly Detection, Feature Engineering, Hyperparameter Tuning (Optuna, Ray Tune)

GenAI & LLMs: GPT-4, Claude API, Llama, LangChain, LlamaIndex, Hugging Face Transformers, Fine-tuning (LoRA, QLoRA), Prompt Engineering, RAG, Vector Embeddings, Agent Frameworks

Computer Vision: OpenCV, Image Classification, Object Detection (YOLO, Faster R-CNN), Semantic Segmentation, Transfer Learning

NLP: NLTK, spaCy, Named Entity Recognition, Sentiment Analysis, Text Classification, Word Embeddings (Word2Vec, GloVe), Sentence Transformers

MLOps & Deployment: MLflow, Weights & Biases, DVC, Docker, Kubernetes, CI/CD (GitHub Actions), Model Monitoring, Drift Detection, Model Serving (FastAPI, TorchServe)

Cloud Platforms: AWS SageMaker, Azure ML, Databricks, Google Cloud AI Platform

Programming: Python (NumPy, Pandas, Matplotlib, Seaborn), R, C++, C, JavaScript, Git/GitHub, REST APIs, Pytest

Advanced Techniques: Transfer Learning, Domain Adaptation, Contrastive Learning, Self-Supervised Learning, Reinforcement Learning, Few-Shot Learning

Vector Databases: Pinecone, ChromaDB, Weaviate, FAISS

Model Optimization: Quantization, Knowledge Distillation, Mixed Precision Training, ONNX, TensorRT

Publications & Research

Investigating Subjective Motor Activity Perception and Gait in Parkinson's Disease

Slim, S., Küderle, A., Moradi, H., **Mittal, M.**, Salin, E., Winkler, J., & Eskofier, B.

IEEE-EMBS International Conference on Biomedical and Health Informatics 2025

Languages

English (Fluent) | German (Elementary - A2) | Hindi (Native)